

BUSINESS INTELLIGENCE & CUSTOMER ANALYTICS SYSTEM

Complete End-to-End Project Report

Excel • MySQL / SQL • Power BI • DAX

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Portfolio / Data Analytics Project

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1. Executive Project Summary

The Business Intelligence & Customer Analytics System is an end-to-end analytics solution that transforms structured business transaction data into actionable business intelligence. The workflow covers Excel preparation and validation, MySQL database management, SQL analysis, Power BI data modeling, DAX calculations, interactive dashboards, business insights and recommended actions.

The solution analyzes business performance, customers, products, geography, transaction status, sales channels, payment methods and time-based trends. Customer analytics is extended through purchase-frequency analysis, customer revenue segmentation and RFM (Recency, Frequency, Monetary) segmentation.

The final Power BI report contains five analytical pages: Executive Overview, Customer Analytics, Product/Service Analytics, Regional Analytics, and Time & Trend Analysis.

2. Business Problem

- Large transaction data is difficult to analyze manually.
- Management needs centralized visibility into revenue, cost and profitability.
- Customer purchasing behavior and customer value need structured analysis.
- Product and category performance need comparison.
- Regional performance needs comparison at region, state and city levels.
- Time-based trends and growth need monitoring.
- Returns and cancellations require operational visibility.
- Business findings need to be translated into recommended actions.

3. Project Objectives

3.1 Business Objectives

- Measure revenue, cost, profit and profitability.
- Understand customer behavior and customer value.
- Identify high-performing products and categories.
- Compare regional, state and city performance.
- Analyze monthly, quarterly and yearly trends.
- Segment customers using RFM.
- Identify risks and opportunities.
- Provide evidence-based recommended business actions.

3.2 Technical Objectives

- Prepare and validate data in Excel.
- Design and populate a relational MySQL database.
- Perform business analysis using SQL.
- Build a Power BI star-schema model.
- Develop reusable DAX measures.
- Build an interactive five-page dashboard.
- Reconcile SQL and Power BI results.
- Demonstrate a refreshable reporting workflow.

4. Business Requirements & Questions

Area	Key Questions
Business Performance	How is the business performing? What are revenue, cost, profit, margin and AOV?
Customer Analytics	Who are the customers? How frequently do they purchase? Which customers need attention?
Product Analytics	Which categories/products generate the strongest revenue, profit and margins?
Regional Analytics	Which regions, states and cities perform best?
Time & Trend Analysis	How does performance change over time? What are the strongest periods and growth trends?

5. Technology Stack

Technology	Purpose
Excel	Data preparation, calculations and validation
MySQL	Relational database and storage
SQL	Import, validation, business analysis and RFM
Power BI	Data modeling and interactive reporting
DAX	KPI, time, customer and RFM calculations

6. Dataset Overview

Final dataset period: January 2024 – December 2026.

Table	Records
dim_customer	7,500
dim_product	150
dim_geography	121
dim_date	1,096
fact_transaction	100,000

7. Data Architecture & Data Model

Raw Business Data



Excel Preparation & Validation



CSV / Structured Data



MySQL Database



SQL Validation & Business Analysis



Power BI Data Model



DAX



Interactive Dashboard



Business Insights & Actions



Figure 1. System Architecture of the Business Intelligence & Customer Analytics System

7.1 Star Schema

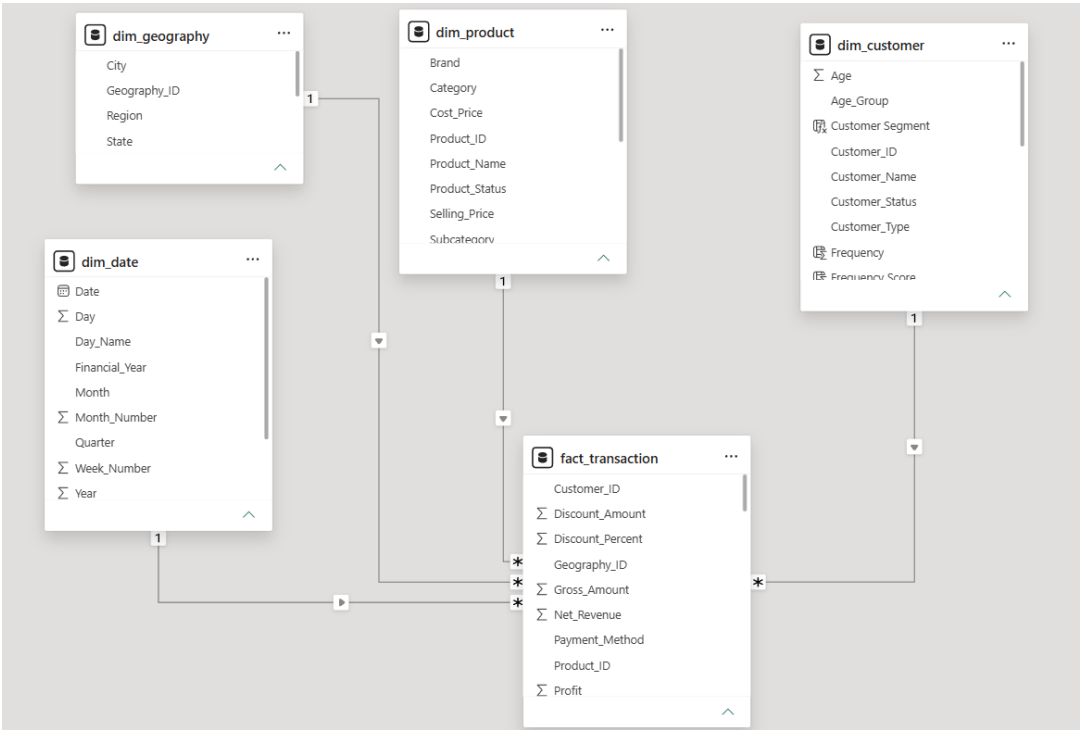
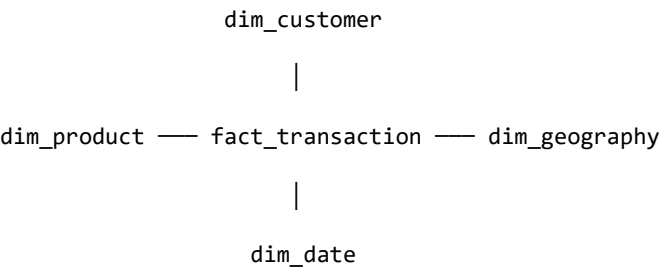


Figure 2. Power BI Star Schema showing Dimension and Fact Table Relationships

7.2 Relationships

Dimension	Fact	Cardinality	Direction
dim_customer	fact_transaction	1 : Many	Single
dim_product	fact_transaction	1 : Many	Single
dim_geography	fact_transaction	1 : Many	Single
dim_date	fact_transaction	1 : Many	Single

7.3 Main Fields

Table	Important Fields
dim_customer	Customer_ID, Customer_Name, Gender, Age, Age_Group, Join_Date, Customer_Type, Geography_ID, Customer_Status
dim_product	Product_ID, Product_Name, Category, Subcategory, Brand, Cost_Price, Selling_Price, Product_Status
dim_geography	Geography_ID, City, State, Region
dim_date	Date, Day, Day_Name, Week_Number, Month, Month_Number, Quarter, Year, Year_Month, Financial_Year
fact_transaction	Transaction_ID, Transaction_Date, Customer_ID, Product_ID, Geography_ID, Quantity, Unit_Price, Discount_Percent, Discount_Amount, Gross_Amount, Net_Revenue, Unit_Cost, Total_Cost, Profit, Profit_Margin, Payment_Method, Sales_Channel, Transaction_Status

8. Excel Data Preparation

The initial preparation and inspection layer was Excel. The working sheets were converted into structured Excel Tables.

Excel Table	Purpose
tbl_customer	Customer master
tbl_product	Product master
tbl_geography	Geography master
tbl_date	Date dimension
tbl_transaction	Transaction fact

- Date formatting and type checks
- Financial/accounting formatting
- Percentage formatting
- ID consistency
- Duplicate checks
- Missing-value checks
- Calculation checks
- Reference validation

	A	B	C	D	E	F	G	H	I	J
1	Transaction_ID	Transaction_Date	Customer_ID	Product_ID	Geography_ID	Quantity	Unit_Price	Discount_Percent	Discount_Amount	Gross_Amount
2	TXN00000001	2026-12-10	CUST06831	PROD0068	GEO108	1	757.30	0.15	113.59	
3	TXN00000002	2025-08-25	CUST00017	PROD0044	GEO085	1	2393.73	0.05	119.69	2
4	TXN00000003	2026-02-06	CUST01764	PROD0063	GEO071	1	322.05	0.00	0.00	
5	TXN00000004	2026-10-27	CUST02625	PROD0074	GEO118	1	3178.98	0.00	0.00	3
6	TXN00000005	2024-02-04	CUST00020	PROD0040	GEO032	1	736.15	0.02	14.72	
7	TXN00000006	2026-12-09	CUST01926	PROD0028	GEO044	1	3210.78	0.10	321.08	3
8	TXN00000007	2024-12-08	CUST02517	PROD0133	GEO055	1	5300.00	0.15	795.00	5
9	TXN00000008	2024-05-06	CUST07472	PROD0050	GEO053	1	4202.39	0.05	210.12	4
10	TXN00000009	2026-03-02	CUST00257	PROD0149	GEO021	1	1906.09	0.15	285.91	1
11	TXN00000010	2024-11-16	CUST03100	PROD0009	GEO078	4	1874.74	0.10	749.90	7
12	TXN00000011	2024-03-26	CUST04901	PROD0018	GEO116	1	1991.83	0.15	298.77	1
13	TXN00000012	2025-06-10	CUST06732	PROD0084	GEO092	5	2063.89	0.08	825.56	10
14	TXN00000013	2026-04-13	CUST06004	PROD0078	GEO061	1	1193.30	0.15	178.99	1
15	TXN00000014	2024-08-15	CUST05636	PROD0116	GEO023	1	361.17	0.05	18.06	
16	TXN00000015	2025-03-09	CUST06115	PROD0149	GEO102	2	1906.09	0.00	0.00	3
17	TXN00000016	2024-12-21	CUST01762	PROD0038	GEO003	3	1599.30	0.10	479.79	4
18	TXN00000017	2025-08-15	CUST06195	PROD0137	GEO101	1	3492.21	0.05	174.61	3
19	TXN00000018	2025-12-02	CUST05641	PROD0079	GEO056	2	2652.20	0.00	0.00	5
20	TXN00000019	2024-07-24	CUST00229	PROD0040	GEO020	3	736.15	0.02	44.17	2
21	TXN00000020	2025-08-24	CUST06200	PROD0145	GEO108	1	2018.03	0.02	40.36	2
22	TXN00000021	2025-05-24	CUST01762	PROD0035	GEO003	1	845.05	0.08	67.60	
23	TXN00000022	2026-05-04	CUST02860	PROD0100	GEO102	2	3145.70	0.00	0.00	6
24	TXN00000023	2026-01-17	CUST06016	PROD0107	GEO052	2	1557.85	0.08	249.26	3

< >

Dim_CustomerDim_ProductDim_GeographyDim_DateFact_Transaction

+ : ◀ ▶

Figure 3. Prepared Excel Transaction Data Structure

9. MySQL Database

```
CREATE SCHEMA Project1_BusinessAnalytics;
```

```
USE Project1_BusinessAnalytics;
```

9.1 Tables

- dim_customer
- dim_product
- dim_geography
- dim_date
- fact_transaction

9.2 Data Types

Type	Use
VARCHAR	IDs and descriptive fields
INT	Age, quantity and date-part numbers
DATE	Join/transaction/calendar dates
DECIMAL	Financial and precise percentage values

```
-- Example financial storage
```

```
DECIMAL(12,2)
```

```
DECIMAL(14,2)
```

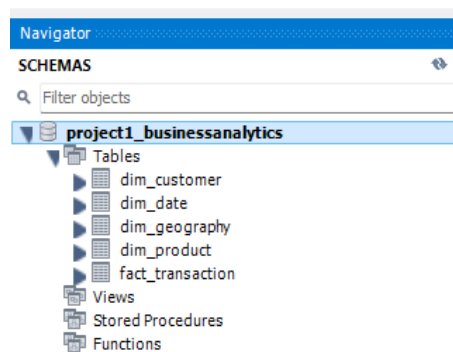


Figure 4. MySQL Database Schema and Project Tables

10. SQL Data Import & Validation

- Excel → CSV preparation
- Date-format correction for date CSVs
- MySQL table creation
- CSV import / Table Data Import Wizard
- Row-count verification
- Integrity and calculation validation

Table	Rows
dim_customer	7,500
dim_product	150
dim_geography	121
dim_date	1,096
fact_transaction	100,000

```

1 • SELECT 'Customers' AS Validation_Check, COUNT(*) AS Actual_Count
2   FROM dim_customer
3   UNION ALL
4   SELECT 'Products', COUNT(*)
5   FROM dim_product
6   UNION ALL
7   SELECT 'Geographies', COUNT(*)
8   FROM dim_geography
9   UNION ALL
10  SELECT 'Dates', COUNT(*)
11  FROM dim_date
12  UNION ALL
13  SELECT 'Transactions', COUNT(*)
14  FROM fact_transaction;
15

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
Validation_Check	Actual_Count				
Customers	7500				
Products	150				
Geographies	121				
Dates	1096				
Transactions	100000				

Figure 5. SQL Data Validation Results

10.1 Financial Reconciliation

Metric	Result
Revenue	₹412,829,960.96
Cost	₹326,765,369.15
Profit	₹86,064,591.81
Average Order Value	₹4,128.30
Overall Margin	20.85%

11. SQL Business Analysis

- Overall business performance
- Transaction status
- Customer behavior and value
- Purchase frequency
- Customer revenue segmentation
- RFM segmentation
- Product/category performance
- Region/state/city performance
- Sales channel
- Payment method
- Monthly, quarterly and yearly trends
- MoM and YoY growth

11.1 Representative Management Query

SELECT

```
COUNT(*) AS Total_Transactions,  
  
COUNT(DISTINCT Customer_ID) AS Unique_Customers,  
  
ROUND(SUM(Net_Revenue), 2) AS Total_Revenue,  
  
ROUND(SUM(Total_Cost), 2) AS Total_Cost,  
  
ROUND(SUM(Profit), 2) AS Total_Profit,  
  
ROUND(AVG(Net_Revenue), 2) AS Average_Order_Value,  
  
ROUND((SUM(Profit) / SUM(Net_Revenue)) * 100, 2)  
  
AS Overall_Profit_Margin
```

FROM fact_transaction;

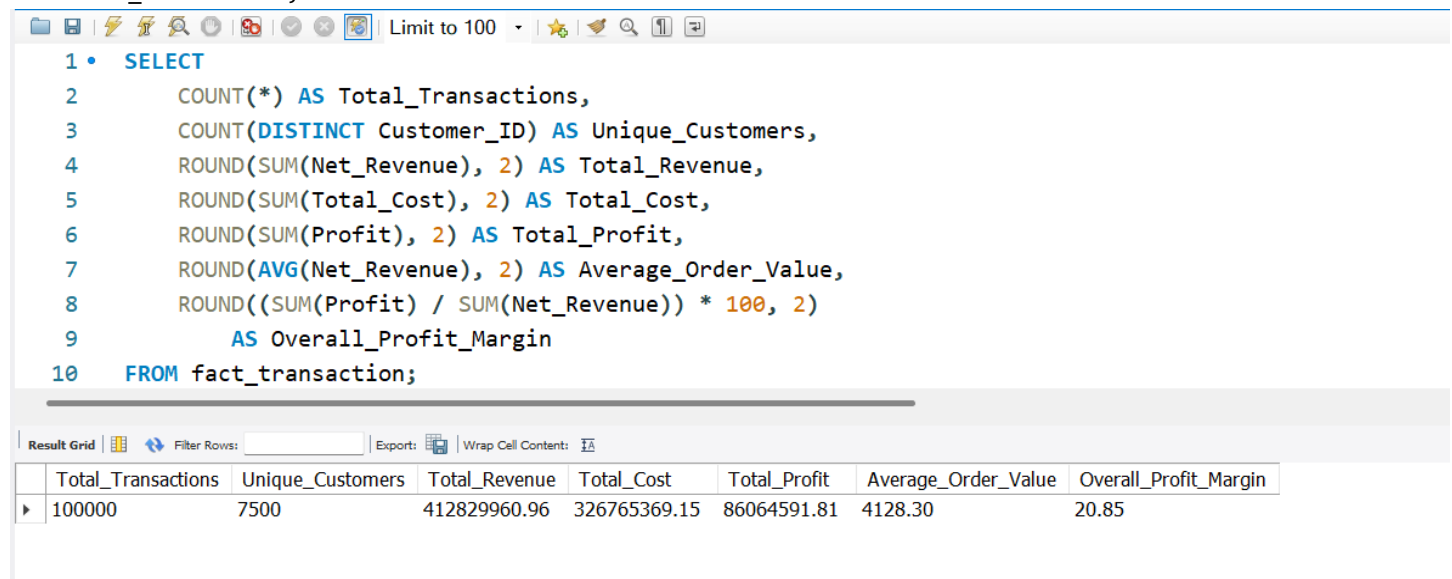


Figure 6. SQL Business Analysis – Management Summary Results

12. Customer & RFM Analytics

Metric	Result
Unique Customers	7,500
New Customers	3,754
Existing Customers	3,746
Repeat Customers	7,500 (modeled dataset)
Revenue per Customer	₹55,043.99
Transactions per Customer	13.33

The repeat-customer result reflects the modeled dataset and indicates recurring transaction activity during the analysis period.

12.1 Purchase Frequency

Band	Customers
2–5	63
6–10	1,635
11–20	5,569
20+	233

12.2 Customer Revenue Segmentation

Segment	Customers
Below 25K	355
25K–50K	2,936
50K–100K	4,020
100K–150K	187
Above 150K	2

12.3 RFM Method

- Recency: how recently a customer purchased.
- Frequency: how frequently a customer purchased.
- Monetary: how much revenue a customer generated.
- Deterministic ranking uses Customer_ID as a tie-breaker so SQL and Power BI produce matching buckets.

12.4 Final RFM Segmentation

Segment	Customers	Share
Loyal Customers	3,321	44.28%
Potential Customers	2,381	31.75%
At Risk	1,185	15.80%
Low Engagement	538	7.17%
Champions	75	1.00%

12.5 RFM Financial Contribution

Segment	Revenue	Profit
Loyal Customers	₹21.33 Cr	₹4.47 Cr
Potential Customers	₹11.71 Cr	₹2.42 Cr
At Risk	₹5.17 Cr	₹1.07 Cr
Low Engagement	₹2.41 Cr	₹0.50 Cr
Champions	₹0.67 Cr	₹0.14 Cr

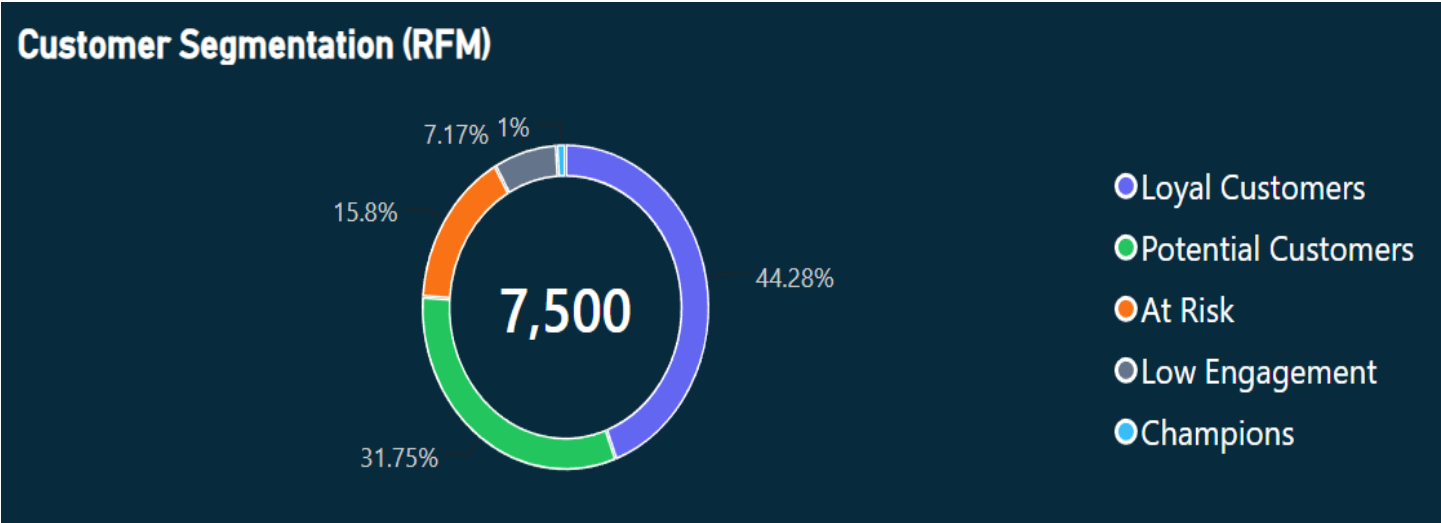


Figure 7. RFM Customer Segmentation Analysis

13. Power BI Data Connection & Model

The validated MySQL database was connected to Power BI Desktop. The five project tables were loaded and relationships were configured as active 1-to-many relationships from dimensions to fact_transaction.

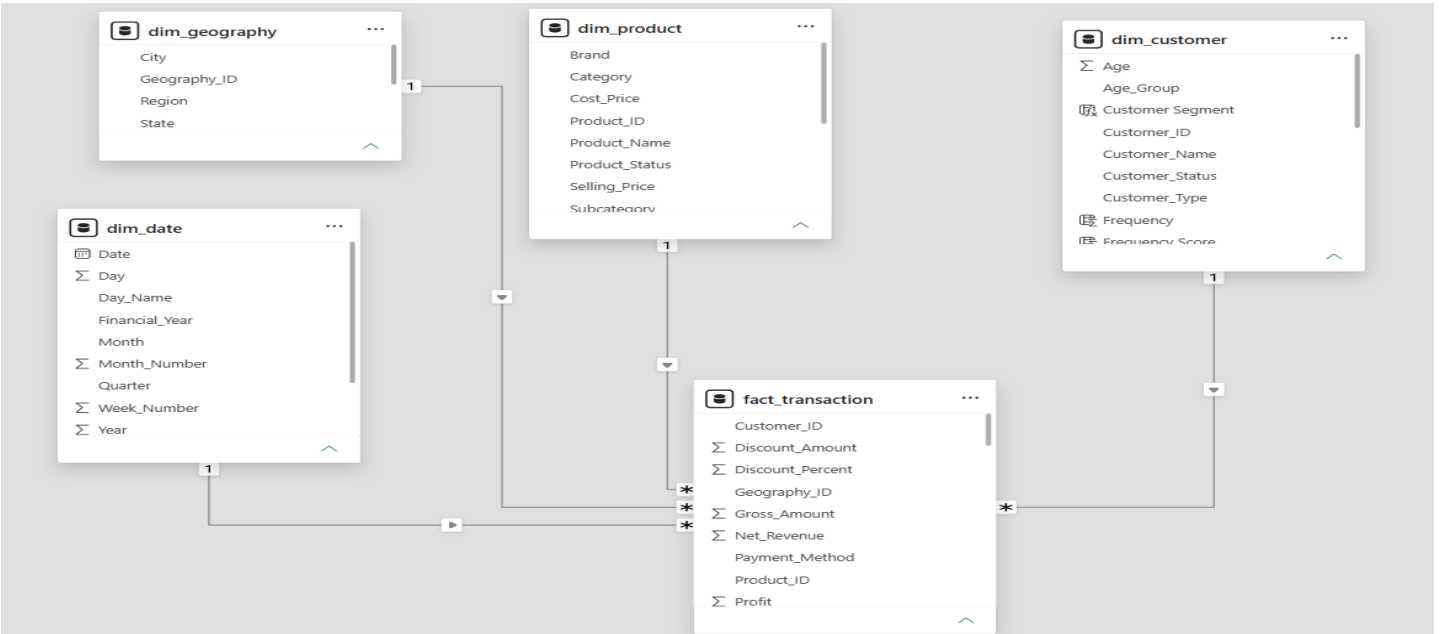


Figure 8. Power BI Data Model and Table Relationships

14. DAX Measures & Calculations

14.1 Core KPI Measures

Total Revenue =

```
SUM(fact_transaction[Net_Revenue])
```

Total Profit =

```
SUM(fact_transaction[Profit])
```

Total Transactions =

```
COUNTROWS(fact_transaction)
```

Average Order Value =

```
AVERAGE(fact_transaction[Net_Revenue])
```

Overall Profit Margin =

```
DIVIDE([Total Profit], [Total Revenue], 0)
```

Unique Customers =

```
DISTINCTCOUNT(fact_transaction[Customer_ID])
```

14.2 Customer Measures

Revenue per Customer =

```
DIVIDE([Total Revenue], [Unique Customers], 0)
```

Profit per Customer =

```
DIVIDE([Total Profit], [Unique Customers], 0)
```

Transactions per Customer =

```
DIVIDE([Total Transactions], [Unique Customers], 0)
```

14.3 Transaction Measures

Returned Transactions =

```
CALCULATE(
```

```
    COUNTROWS(fact_transaction),
```

```

fact_transaction[Transaction_Status] = "Returned"
)

```

```

Cancelled Transactions =
CALCULATE(
    COUNTROWS(fact_transaction),
    fact_transaction[Transaction_Status] = "Cancelled"
)

```

```

Return Rate =
DIVIDE([Returned Transactions], [Total Transactions], 0)

```

```

Cancellation Rate =
DIVIDE([Cancelled Transactions], [Total Transactions], 0)

```

14.4 Time Intelligence

```

YTD Revenue =
TOTALYTD([Total Revenue], dim_date[Date])

```

```

Revenue YoY % =

```

```

VAR PreviousYearRevenue =
    CALCULATE(
        [Total Revenue],
        DATEADD(dim_date[Date], -1, YEAR)
    )
RETURN
    DIVIDE(
        [Total Revenue] - PreviousYearRevenue,
        PreviousYearRevenue,
        0
    ) * 100

```

14.5 RFM DAX

- Recency, Frequency and Monetary customer columns.
- Deterministic scoring with Customer_ID tie-breaker.
- RFM Score combining the three component scores.
- Customer Segment assignment.

```
1 Customer Segment =  
2 SWITCH(  
3     TRUE(),  
4     dim_customer[RFM Score] = "555", "Champions",  
5     dim_customer[Recency Score] = 5  
6     || dim_customer[Frequency Score] = 5  
7     || dim_customer[Monetary Score] = 5, "Loyal Customers",  
8     dim_customer[Recency Score] = 3  
9     || dim_customer[Frequency Score] = 3  
10    || dim_customer[Monetary Score] = 3, "Potential Customers",  
11    dim_customer[Recency Score] = 2  
12    || dim_customer[Frequency Score] = 2  
13    || dim_customer[Monetary Score] = 2, "At Risk",  
14    "Low Engagement"  
15 )
```

Figure 9A. DAX RFM Customer Segmentation Calculation

```
1 Purchase Frequency =  
2 VAR TransactionCount =  
3     CALCULATE(  
4         COUNTROWS(fact_transaction),  
5         RELATEDTABLE(fact_transaction)  
6     )  
7 RETURN  
8     SWITCH(  
9         TRUE(),  
10        TransactionCount <= 5, "2-5",  
11        TransactionCount <= 10, "6-10",  
12        TransactionCount <= 20, "11-20",  
13        "20+"  
14    )
```

Figure 9B. DAX RFM Purchase Frequency Calculation

15. Power BI Dashboard

The dashboard uses a consistent professional dark theme with sidebar navigation, KPI cards, interactive slicers, analytical visuals and insight/action cards.

15.1 Executive Overview

KPIs

- Total Transactions
- Total Revenue
- Total Cost
- Total Profit
- AOV
- Overall Profit Margin

Visuals

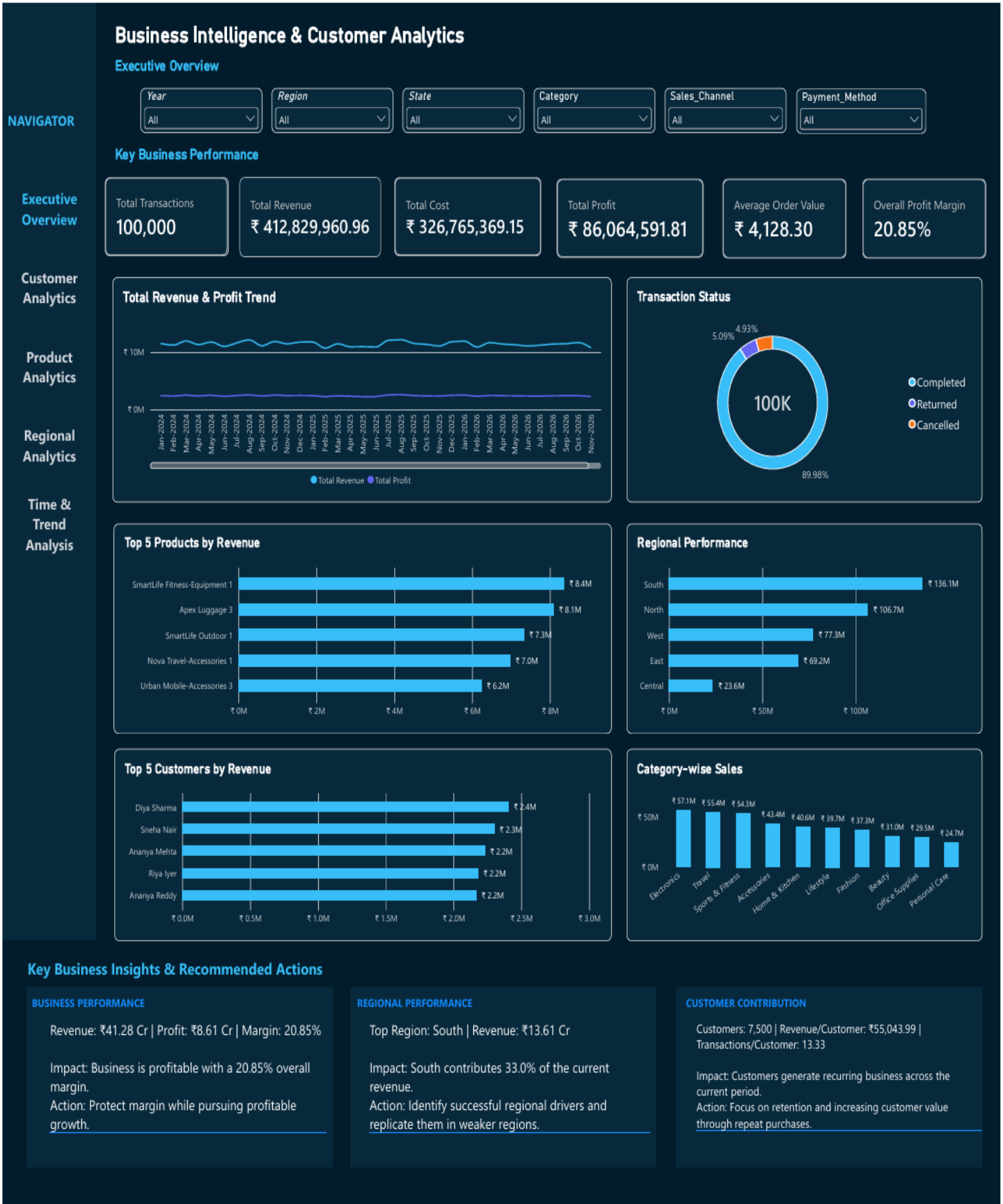
- Monthly Revenue & Profit
- Transaction Status
- Top 5 Products
- Regional Performance
- Top 5 Customers
- Category Sales

Key Insights

- Total revenue is **₹41.28 Cr**, with total profit of **₹8.61 Cr**.
- Overall profit margin is **20.85%**.
- South is the leading region, contributing **₹13.61 Cr** in revenue.
- Customers contribute recurring business with **7,500 customers** and **13.33 transactions per customer**.

Recommended Actions

- Protect the current profit margin while pursuing profitable growth.
- Identify the drivers behind South's strong performance and replicate them in weaker regions.
- Strengthen customer retention and increase customer value through repeat purchases.



15.2 Customer Analytics

KPIs

- Unique Customers
- New Customers
- Existing Customers
- Repeat Customers
- Revenue per Customer
- Transactions per Customer

Visuals

- New vs Existing
- Purchase Frequency
- Customer Revenue Segments
- Top 10 Customers Revenue
- Top 10 Customers Profit
- RFM Segmentation
- Segment Revenue & Profit

Key Insights

- The business has **7,500 unique customers**.
- Revenue per customer is **₹55,043.99**.
- Average transactions per customer are **13.33**.
- **3,321 customers (44.28%)** are classified as Loyal Customers.
- **2,381 customers (31.75%)** are Potential Customers.
- Champions represent **75 customers (1%)**.

Recommended Actions

- Focus on retaining the large Loyal Customer segment.
- Convert Potential Customers into Loyal Customers through targeted engagement.
- Develop reactivation strategies for At Risk customers.
- Increase customer value through repeat purchases and personalized offers.



Figure 11. Customer Analytics Dashboard

15.3 Product/Service Analytics

KPIs

- Total Products
- Active Products
- Revenue
- Profit
- Average Selling Price
- Average Product Margin

Visuals

- Category Revenue
- Category Profit
- Top 10 Products Revenue
- Top 10 Products Profit
- Product Margin
- Subcategory Performance

Key Insights

- The business contains **150 products**, all currently active.
- Total revenue is **₹41.28 Cr** and total profit is **₹8.61 Cr**.
- Average selling price is **₹2,395.12**.
- Average product margin is **24.39%**.
- Electronics is the leading category by revenue at approximately **₹5.71 Cr**.

Recommended Actions

- Protect and expand high-performing product categories.
- Prioritize products with strong profitability rather than revenue alone.
- Review low-performing products and subcategories for improvement opportunities.
- Maintain a healthy product mix with strong-margin products.

15.4 Regional Analytics

KPIs

- Total Regions
- Total States
- Total Cities
- Revenue
- Profit
- Margin

Visuals

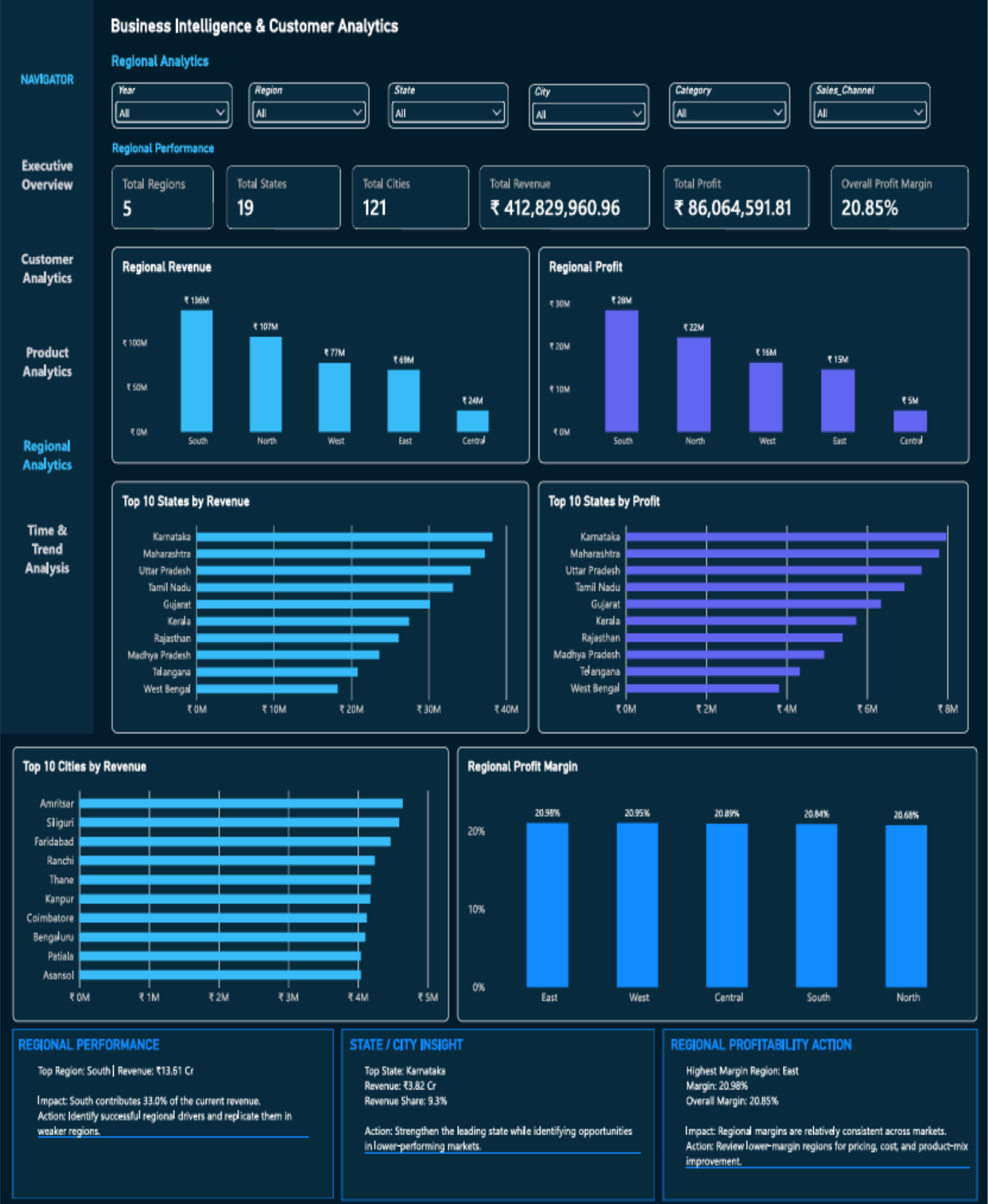
- Regional Revenue
- Regional Profit
- Top 10 States Revenue
- Top 10 States Profit
- Top 10 Cities Revenue
- Regional Margin

Key Insights

- The business operates across **5 regions, 19 states and 121 cities**.
- South is the top region with **₹13.61 Cr revenue**, contributing approximately **33%** of total revenue.
- Karnataka is the leading state with **₹3.82 Cr revenue**.
- East records the highest regional profit margin at approximately **20.98%**.

Recommended Actions

- Replicate the successful drivers of South in weaker regions.
- Strengthen operations and customer acquisition in high-potential states.
- Investigate regional margin differences to identify efficiency opportunities.



15.5 Time & Trend Analysis

KPIs

- Revenue
- Profit
- Revenue Growth
- Profit Growth
- YTD Revenue
- YTD Profit

Visuals

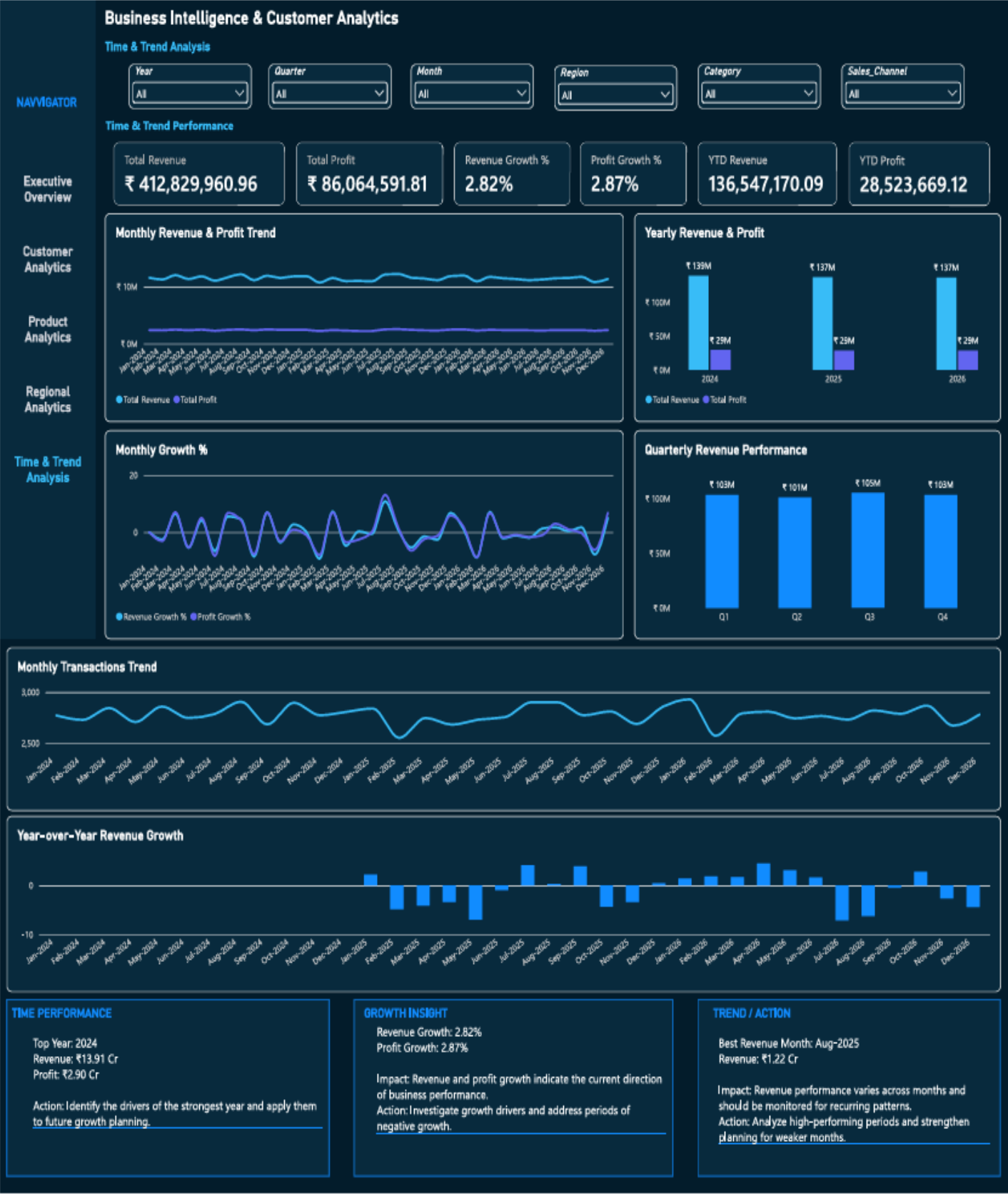
- Monthly Revenue & Profit
- Monthly Growth
- Yearly Revenue & Profit
- Monthly Transactions
- Quarterly Revenue
- YoY Revenue Growth

Key Insights

- **2024** was the strongest year with approximately **₹13.91 Cr revenue** and **₹2.90 Cr profit**.
- Current revenue growth is approximately **2.82%**.
- Current profit growth is approximately **2.87%**.
- **August 2025** was the strongest month with approximately **₹1.22 Cr revenue**.

Recommended Actions

- Monitor monthly and quarterly trends to identify seasonal patterns.
- Investigate the drivers behind high-performing periods.
- Use growth trends to support future sales and resource planning.
- Maintain focus on profitable growth rather than revenue growth alone.



16. Dashboard Interactivity

- Slicers
- Cross-filtering
- Page navigation
- Drill-down
- Drill-through
- Tooltips
- Dynamic DAX calculations

16.1 Major Slicers

Year, Region, State, City, Category, Subcategory, Brand, Sales Channel, Payment Method, Customer Type, Gender, Age Group, Quarter and Month, as applicable to each page.

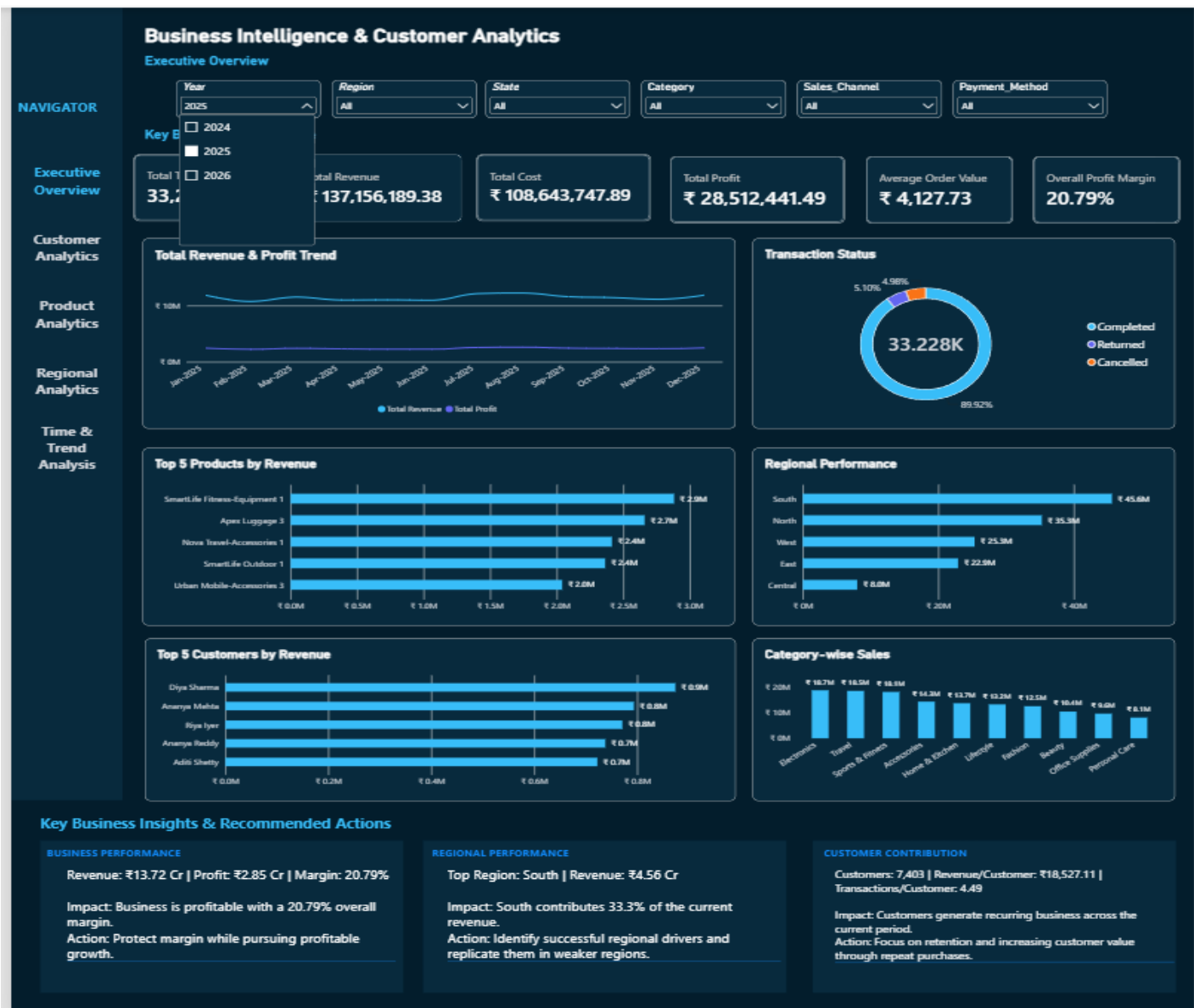


Figure 15. Power BI Dashboard Interactivity with Active Filters

17. Business Insights

Area	Observation	Impact	Recommended Action
Overall Performance	₹41.28 Cr revenue; ₹8.61 Cr profit; 20.85% margin.	Modeled business is profitable.	Protect margin while pursuing profitable growth.
Customer Behavior	7,500 customers; ₹55,043.99 revenue/customer; 13.33 transactions/customer.	Recurring activity is present in the modeled dataset.	Strengthen retention and increase customer value.
RFM	Loyal 3,321 (44.28%); At Risk 1,185 (15.80%).	Segments have different engagement and financial contribution.	Retain Loyal/Champions; convert Potential; re-engage At Risk/Low Engagement.
Product	Electronics is top revenue category at ₹5.71 Cr (13.8%).	Important revenue contributor.	Identify successful drivers and replicate where appropriate.
Region	South ₹13.61 Cr, about 33.0% of revenue; Karnataka ₹3.82 Cr.	Leading markets contribute meaningful revenue.	Identify successful drivers and evaluate weaker markets.
Time	2024 revenue ₹13.91 Cr; August 2025 about ₹1.22 Cr.	Performance varies across periods.	Investigate strong-period drivers and use them for planning.
Transaction Quality	5.09% returned; 4.93% cancelled.	Operational areas require investigation.	Analyze drivers by product, channel and geography.

18. Recommended Business Actions

Area	Focus	Recommended Action	Expected Benefit
Customer Retention	Loyal/Champions	Retention offers and loyalty engagement	Protect recurring revenue
Customer Re-engagement	At Risk/Low Engagement	Targeted campaigns and activity investigation	Recover customer activity
Potential Customers	Large Potential segment	Targeted offers to increase frequency/value	Move customers toward stronger-value segments
Product Strategy	High-performing products	Prioritize profitable products and optimize mix	Support profitable growth
Regional Strategy	Leading regions/states	Identify and replicate successful drivers	Expand successful practices
Operations	Returns/cancellations	Analyze drivers by product/channel/geography	Improve transaction quality
Time Planning	Strong/weak periods	Investigate period drivers and plan resources	Improve planning

Decision framework: Finding → Business Impact → Investigation Area → Recommended Action → Expected Benefit.

19. End-to-End Validation

Layer	Checks	Result
Excel	IDs, missing values, references, calculations	Passed
MySQL	Row counts, PKs, NULLs, references, calculations, date range	Passed
SQL → Power BI	Revenue, cost, profit, transactions, customers	Reconciled
Power BI	Relationships, DAX, KPIs, slicers, charts, navigation, refresh	Passed

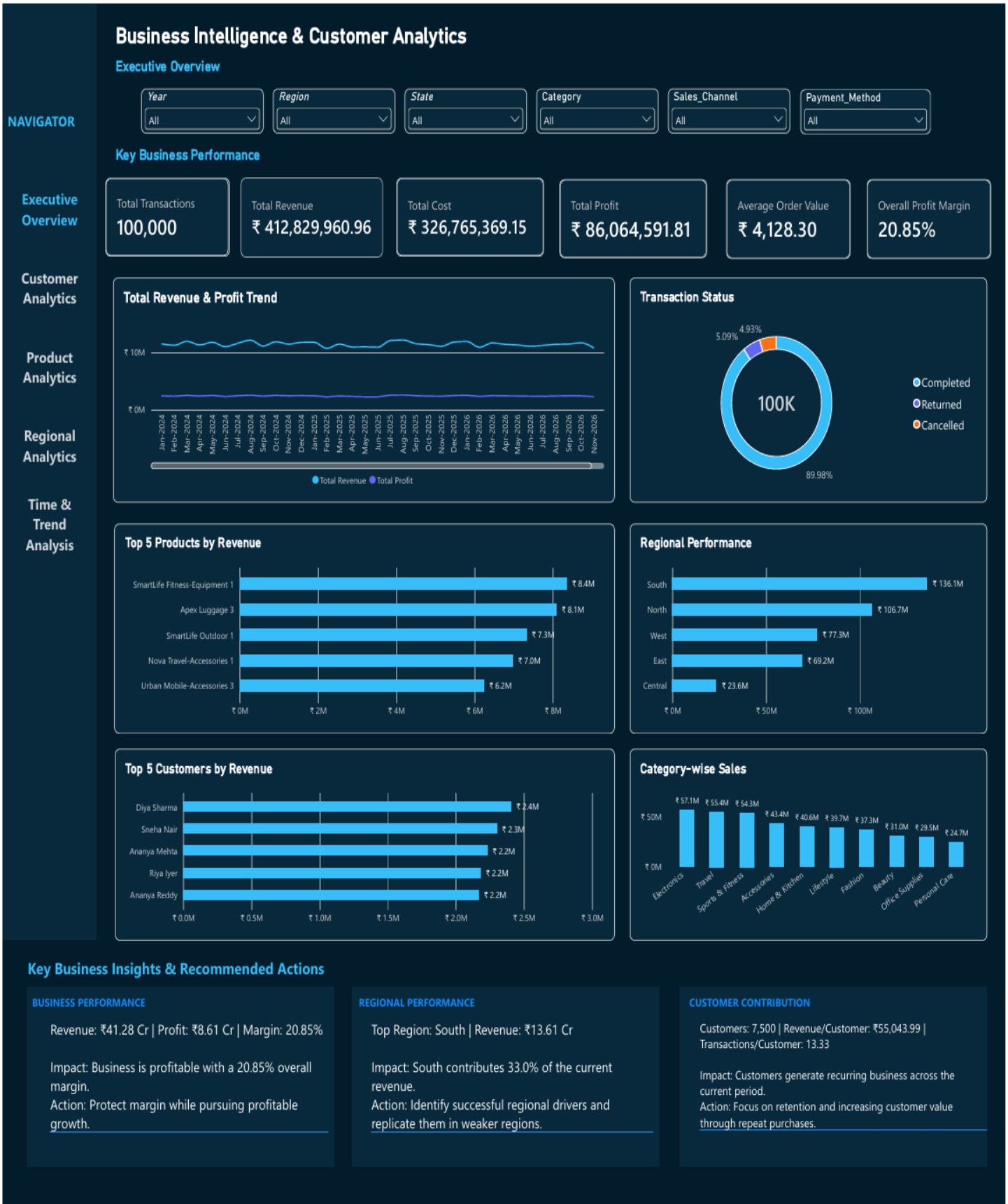


Figure 16. Final Power BI KPI Validation Against SQL Baseline

20. Refresh & Reusability

Updated Source Data



Data Source / MySQL



Power BI Refresh



Model + DAX Recalculation



Updated Dashboard

The solution is designed as a reusable reporting workflow rather than a one-time static analysis.

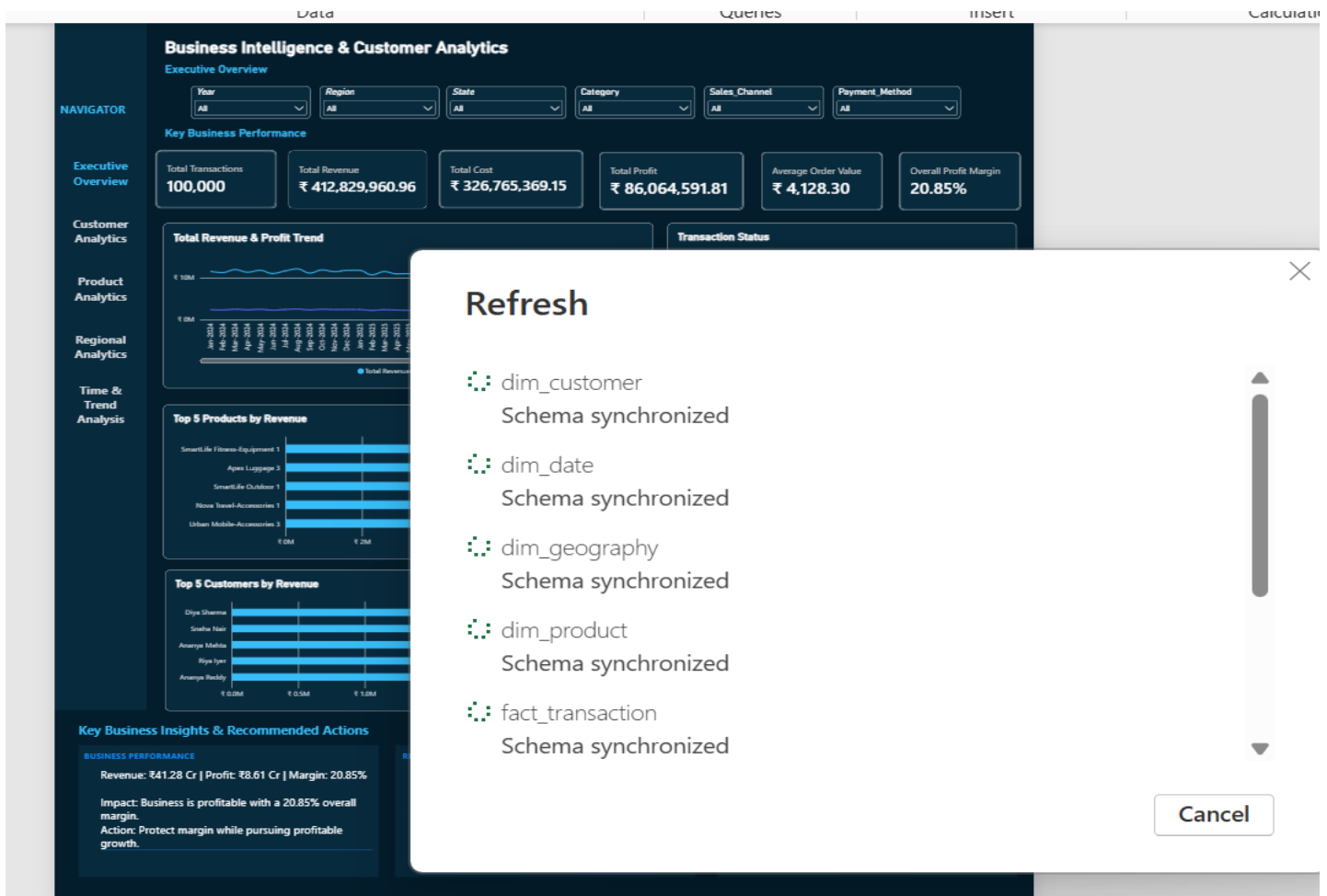


Figure 17. Successful Power BI Data Refresh and Schema Synchronization

21. Final Project Results

KPI	Final Value
Transactions	100,000
Customers	7,500
Revenue	₹41.28 Cr
Cost	₹32.68 Cr
Profit	₹8.61 Cr
AOV	₹4,128.30
Profit Margin	20.85%
Products	150
Regions	5
States	19
Cities	121

22. Technical Skills Demonstrated

Area	Skills
Excel	Data preparation, Excel Tables, validation, financial/business calculations
SQL / MySQL	Database creation, table design, data types, import, joins, aggregations, CASE, CTEs, window functions, RFM, validation
Power BI	MySQL connectivity, star schema, relationships, slicers, dashboards, drill-down, drill-through, tooltips
DAX	KPI measures, CALCULATE, DIVIDE, DISTINCTCOUNT, FILTER, RELATEDTABLE, DATEADD, DATESMTD, TOTALYTD, RFM and dynamic insights
Business Analysis	Performance, customer, product, regional, trend analysis and recommended actions

23. Conclusion

The **Business Intelligence & Customer Analytics System** successfully demonstrates an end-to-end business intelligence and data analytics workflow. The project transforms structured business transaction data into meaningful insights using **Excel, MySQL/SQL, Power BI, and DAX**.

The solution covers business performance, customer behavior, product performance, regional analysis, and time-based trends. Customer analytics is further enhanced through purchase-frequency analysis, customer revenue segmentation, and **RFM customer segmentation**.

The final Power BI solution provides an interactive five-page dashboard with KPIs, analytical visuals, slicers, navigation, and business insight/action sections. SQL and Power BI results were reconciled to validate the accuracy of the reporting outputs, and the refresh workflow was successfully tested.

Overall, the project demonstrates practical skills in **data preparation, SQL analysis, relational data modeling, Power BI dashboard development, DAX calculations, customer analytics, business analysis, and evidence-based decision-making**. It provides a reusable foundation for transforming business data into actionable information for management.